

LOONIBHA TAMRAKAR

–B.tech in AI Student–

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Machine Learning enthusiast with hands-on experience in building end-to-end AI systems across natural language processing and deep learning. Worked on tasks involving semantic search, text classification, and sequence modeling, with a focus on practical implementation and evaluation.

EDUCATION

KATHMANDU UNIVERSITY

Bachelor of Technology in Artificial Intelligence

Dhulikhel, Kavre

Nov 2023 - Expected 2027

Relevant Coursework: Information Retrieval, Pattern Recognition; Advanced Machine Learning; Software Engineering

SKILLS

Programming and System: Python, C++, JS, React, SQL, Git, SDLC, Docker, Streamlit, FastAPI, Django, Flask, Numpy, Pandas

Machine Learning: Tensorflow, Keras, PyTorch, Computer Vision, Neural Networks, NLP, RAG, Vector Databases (FAISS, Chroma), Embeddings, Retrieval Pipelines, Hugging face, Transformers

PROJECTS

Retrieval-Augmented Question Answering & Quiz Generation System

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** system for answering and generating quizzes from textbooks
- Implemented **semantic search** using **SentenceTransformers embeddings** and **FAISS** vector database
- Integrated transformer-based language model (**Flan-T5**) for context-aware answer generation

Yoga Poses Video Classification (CNN + LSTM)

- Implemented a spatio-temporal classification model combining **ResNet based feature extraction** with **LSTM sequence modeling** for exercise recognition from video data
- Explored **transfer learning** to leverage pre-trained visual representations and improve performance on limited labeled data
- Built a lightweight **Streamlit** interface for real-time inference and visualization of predicted exercise classes

Mamacare- Pregnancy Care Application

- Built using **React** (frontend), **FastAPI** (backend), and **SQL** database. Integrated a nutrition recommendation module using Python-based logic and deployed via Streamlit for rapid prototyping.
- Implemented modules including a **nutrition calculator and food recommendation system**, along with tools for partner integration and pregnancy event tracking

EXPERIENCE AND EXTRACURRICULARS

AI FELLOWSHIP — FUSEMACHINES

Apr 2026 - Present

- Selected for a competitive AI fellowship focused on **applied machine learning** and **real-world systems**
- Currently undergoing structured training in **model development, evaluation, and ML workflows**
- Working on hands-on assignments involving data preprocessing, model building, and experimentation

Executive Member — KU AI Club

Feb 2025 – Present

- Contributed to organizing **AI Convergence**, an annual AI focused event with **300+ attendees** bringing together students and practitioners
- Collaborated with peers to promote learning and engagement in artificial intelligence topics

ANNUAL NEPAL AI SCHOOL — NAAMI

Jan 2026

- Completed hands-on labs in **machine learning and deep learning fundamentals**
- Gained practical experience with classification models, neural networks, and evaluation techniques